# Uroš Milutinović

## AI Engineer & Software Developer

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## PROFESSIONAL SUMMARY

AI and software engineer who ships production systems, not just prototypes. Built a RAG pipeline (LangChain, GPT-4, Pinecone) with a FastAPI + React.js service layer handling 500+ queries/day in production; fine-tuned open-source LLMs (Mistral 7B, Llama 3) with LoRA, evaluated through RAGAS and BERTScore pipelines; designed multi-agent and Claude-powered agentic systems, from a voice-controlled Revit BIM plugin to an autonomous job-application agent. B.Sc. in New Computer Technologies (Feb 2026), AWS Machine Learning Specialty certified.

## KEY SKILLS

- **AI / ML:** Python, PyTorch, LangChain, RAG, LLM fine-tuning (LoRA), Hugging Face, OpenAI API, Anthropic Claude API, Pinecone, multi-agent orchestration, evaluation (RAGAS, BERTScore)
- **Full-Stack:** FastAPI, React.js, Java, Spring Boot, PHP, MySQL, PostgreSQL, REST APIs
- **DevOps / Cloud:** Docker, Git, Linux, n8n (workflow automation), AWS (ML Specialty certified), Agile / Scrum
- **Digital & Content:** Meta Ads (Facebook & Instagram), campaign optimization, video editing & publishing, Canva, Adobe Social Media Content & Strategy (certified)

## WORK EXPERIENCE

### AI Junior Engineer — Mar 2026 – Aug 2026

*Netconomy · Belgrade*

- Built, tested, and refined AI/LLM-based applications and workflows as part of a production-oriented engineering team.
- Collaborated cross-functionally to integrate AI technologies into the company's software systems, working within an Agile development process.

### AI & Full-Stack Engineer — Contract / Freelance — Dec 2024 – Present

*Self-employed · Remote (international clients) · alongside B.Sc. studies*

- Designed and shipped a RAG system for document Q&A (LangChain, GPT-4, Pinecone) running at 500+ queries per day, with a FastAPI backend for asynchronous inference and a React.js frontend with response streaming.
- Fine-tuned open-source LLMs (Mistral 7B, Llama 3) on domain-specific data using Hugging Face Transformers and LoRA adapters; added RAGAS / BERTScore evaluation and Docker deployment with structured logging.
- Delivered live client websites, including N Detailing Studio and FitPro Agency — a lead-generation SPA with EmailJS, Google Sheets CRM, Meta Pixel tracking, and an admin panel.
- Run Meta Ads campaigns and produce short-form video content for client marketing funnels.
- Currently building a Claude-powered job-analysis agent and an LLM evaluation framework (see Featured Projects).

### IT Support Engineer — Jun 2024 – Dec 2024

*Microsec d.o.o. · Belgrade*

- Resolved 20+ tickets per week and wrote internal runbooks that cut repeat incidents by 30%.
- Administered MySQL databases with zero data loss; configured developer workstations and supported deployment pipelines and Postman API testing.

## ADDITIONAL EXPERIENCE

**Truck Dispatcher & Track-and-Trace Specialist** — Sky Freight Dispatch (U.S. market), Oct 2025 – May 2026 (part-time). Coordinated U.S. freight transport and GPS tracking; managed all client and driver communication in English.
**Guest Services & Operations — Work and Travel USA Program**, United States, Summer 2025. Full-time guest-facing role in English; resolved a high volume of guest requests in a fast-paced international team.

## FEATURED PROJECTS

**Revit AI Assistant — Voice-Controlled Multi-Agent BIM Automation Plugin**
AI plugin for Autodesk Revit that automates engineers' workflows directly from the 3D building model via natural-language voice commands. Multi-agent architecture: an orchestrator agent interprets intent and delegates to specialized sub-agents for quantity takeoffs, material estimation, and model manipulation.
*Stack: C# / .NET, Revit API, multi-agent LLM orchestration, Speech-to-Text*

**carservice-api — Car Service Workshop REST API (Java/Spring Boot)** · github.com/umilutinovic25-hash/carservice-api
Layered Spring Boot backend for a car-service workshop — customers, vehicles, a service catalog, and work orders driven by a status state machine enforced at the entity level (mechanic-assignment and item rules baked in, not just checked in the service layer), with price-snapshotted line items and invoice generation. MySQL schema versioned with Flyway migrations, H2 profile for fast local runs. 30 tests (JUnit 5, Mockito, full MockMvc integration covering the entire receive-to-paid-invoice lifecycle against a real database).
*Stack: Java 17, Spring Boot 3, Spring Data JPA/Hibernate, MySQL, Flyway, Maven, JUnit 5, Mockito, Docker*

**JobScout — AI Job-Application Agent (Claude API)** · github.com/umilutinovic25-hash/jobscout
Autonomous agent that analyzes job postings against a CV: scrapes the posting, extracts structured requirements with Claude, scores the match on a calibrated 0–100 rubric with gap analysis, drafts a tailored cover letter, and tracks every application — including per-application API cost — in SQLite.
*Stack: Python, Anthropic Claude API, structured outputs, httpx + BeautifulSoup, SQLite, pytest*

**MailGuard — Email Phishing & Fraud Scanner** · github.com/umilutinovic25-hash/mailguard
Security CLI that flags whether inbox emails are safe to open: layered heuristics over full headers — SPF/DKIM/DMARC, brand-spoofing and lookalike domains, masked links, dangerous attachments — plus an LLM second-pass verdict on borderline cases; runs as a scheduled daily scan.
*Stack: Python, IMAP, email authentication (SPF/DKIM/DMARC), LLM triage, launchd*

**Credit Card Fraud Detection — Imbalanced ML + Real-Time Scoring API** · github.com/umilutinovic25-hash/credit-card-fraud-detection
End-to-end fraud detection on 284k card transactions (0.17% fraud): KNN, XGBoost, and LSTM combined in a soft-voting ensemble with SMOTEENN resampling and precision–recall threshold tuning — F1 0.82 on a held-out set with the true fraud distribution. Deployed as a FastAPI scoring service (measured p50 latency 3.5 ms, pytest contract tests) plus an interactive demo simulating a day of bank transactions with an adjustable decision threshold.
*Stack: Python, scikit-learn, XGBoost, PyTorch, imbalanced-learn, FastAPI, Gradio, pytest*

**dine-agent — AI Phone Host for Restaurant Reservations** · github.com/umilutinovic25-hash/dine-agent
Claude-powered voice agent that answers calls and manages the booking book. Every write goes through a policy-owning engine — service windows, table capacities, dining durations, conflict detection — so the model proposes but can never invent a table.
*Stack: Python, Anthropic Claude API (tool use), FastAPI, Web Speech API, pytest*

**llm-eval — LLM Evaluation Framework** · github.com/umilutinovic25-hash/llm-eval
Framework for comparing LLMs (Claude, local models via Ollama) on the same task suite — accuracy, cost, and latency side by side. YAML-defined evals, async runner with per-request token and cost tracking, JSON snapshots for regression detection, non-zero exit codes for CI gating, and a self-contained HTML report dashboard.
*Stack: Python, asyncio, Anthropic API, Ollama, YAML, pytest*

**Semantic Search over 50k ArXiv Papers — NLP Embeddings** · github.com/umilutinovic25-hash/semantic-search-arxiv
Semantic search engine over 50,000 ML paper abstracts: sentence-transformer embeddings with exact cosine ranking answer conceptual queries that share zero keywords with the results (side-by-side TF-IDF comparison included), plus related-paper recommendations and a t-SNE cluster map of the corpus.
*Stack: Python, sentence-transformers, scikit-learn, NumPy, CLI*

**Other projects** · all on github.com/umilutinovic25-hash
poslovi-mcp (Model Context Protocol server for job-board search) · Serbian LLM Benchmark (8 open models evaluated on a Serbian-language task suite, published) · Razumem Nalaz (on-device OCR lab-report explainer PWA, live) · SafeCircle (personal-safety app backend — FastAPI, Twilio) · N Detailing Studio (live client website) · Car Service Platform (B.Sc. thesis — PHP/MySQL, 3NF design)

## CERTIFICATIONS

- AWS Certified Machine Learning – Specialty — AWS, 2026
- Claude Code, AI Fluency, MCP & Claude API — Anthropic Academy; Monitoring Production Agents — LangChain Academy, 2026
- AI Agents (GAIA: 100%), MCP, LLM/NLP & Deep RL — Hugging Face, 2026
- GitHub Foundations, HackerRank SQL (Advanced) & Problem Solving (Intermediate) — 2026
- Feature Engineering, Time Series & Intro to ML — Kaggle Learn; Databricks Fundamentals — Databricks, 2026
- Machine Learning & Advanced Learning Algorithms — DeepLearning.AI, 2026

## LANGUAGES

- English — C1 (fluent business English)
- Serbian — native
- German — A2

## EDUCATION

**B.Sc., New Computer Technologies** **Graduated Feb 2026**
*VIŠER — College of Electrical Engineering and Computer Science, Belgrade (2022 – 2026)*

**Mechatronics Technician** **2018 – 2022**
*Technical School "17. Septembar", Lajkovac*